



BA and BSc Degree Examinations 2022-23
SECURE EXAM

SOLUTION GUIDE

Department:
Management

Title of Exam:
Business Economics

Time Allowed:
THREE hours

Allocation of Marks:
This paper is in two sections: Section A and Section B.
Each section is worth 50 marks.

Instructions for Candidates:
Candidates should answer **ALL** questions.

Candidates may not use red pen and/or pencil to write their answers.

Materials Supplied:
Green Booklet
Calculator

Do not write on this booklet before the exam begins
Do not turn over this page until instructed to do so by an invigilator

Section A (2 marks per question)

Question 1

Tom and Sara are planning to go out in the evening. Tom likes to go to the theatre, and Sara likes to go to the opera. Given the information in the table below, which statement is not correct?

		Sara	
Tom	Theatre	2,1	0,0
	Opera	0,0	1,2

- A Both going to the opera is a Nash equilibrium.
- B Both going to the theatre is a Nash equilibrium.
- C Tom going to the theatre and Sara going to the opera is a Nash equilibrium.**
- D There is at least one Nash equilibrium.

Question 2

Sam only eats Brand A. If there is an increase in Sam's budget, all of which is spent on Brand A, and if prices do not change, which will be true?

- A There will be movement down along the ray of Brand A cereal.
- B There will be movement up along the ray of Brand A cereal.**
- C The ray of Brand A cereal will pivot towards the vertical axis.
- D The ray of Brand A cereal will pivot towards the horizontal axis.
- E none of the above.

Question 3

An oligopolist may engage in short-run behaviour that results in lower profits if

- A it lends credibility to the firm's threats.**
- B it leads to a Nash equilibrium.
- C it is a dominant strategy.
- D it is not involved in a repeated game.
- E none of the above.
- F all of the above.

Question 4

When a firm in monopolistic competition raises its price, it

- A loses all of its customers.
- B loses many, but not all, of its customers.**
- C loses no customers at all.
- D none of the above.

Question 5

A monopolist finds that if it remains open, the best output is 50 a week, but it makes a loss at this output. Under what circumstances should it shut down?

- A If AR at this output is below AC.
- B If AR at this output is below AVC.**
- C If MR at this output is below AC.
- D If MR at this output is below AVC.
- E none of the above.

Question 6

To prevent the value of the Pound from depreciating against the Euro, the Bank of England might:

- A buy the Pound in the foreign exchange market and increase its foreign exchange reserves.
- B buy the Pound in the foreign exchange market and decrease its foreign exchange reserves.**
- C sell the Pound in the foreign exchange market and increase its foreign exchange reserves.
- D sell the Pound in the foreign exchange market and decrease its foreign exchange reserves.
- E none of the above.
- F all of the above.

Question 7

If you were running a firm in a perfectly competitive industry, you would be spending time making decisions on

- A what price to charge.
- B how much of each input to use.**
- C how much to spend on advertising.
- D how to prevent new firms from joining the market.
- E none of the above.
- F all of the above.

Question 8

The normal turnover of labour and the usual time it takes to find a satisfactory job opening causes _____ unemployment to persist even at potential GDP.

- A frictional.**
- B involuntary.
- C structural.
- D cyclical.
- E none of the above.
- F all of the above.

Question 9

A firm that employs almost all the labour force in a particular town is an example of

- A Positive externality
- B Monopsony power**
- C Monopoly power
- D Negative externality
- E Public good
- F none of the above.
- G all of the above.

Question 10

A firm will find it difficult to engage in price discrimination if:

- A there are close substitutes.**
- B there are many consumers.
- C the good is normal.
- D consumers are smart
- E none of the above.
- F all of the above.

Question 11

Assume that the marginal propensity to consume domestically produced goods is 0.7, and there are no taxes. The government decides to increase public spending by £100 million. According to simple Keynesian multiplier analysis, what is the total change in national income resulting from this increase in government expenditure (to the nearest million)?

- A £70 million.
- B £143 million.
- C £170 million.
- D £333 million.**
- E none of the above.

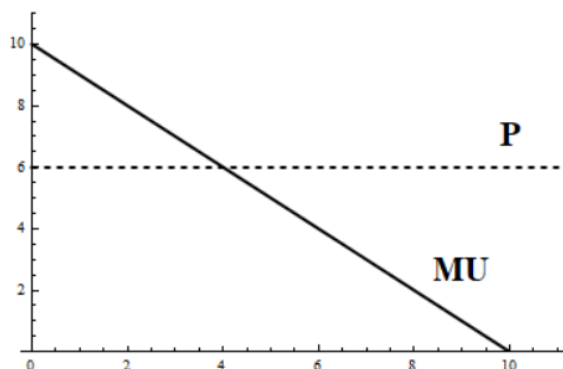
Question 12

One method of intervening in trade is dumping surplus EU wheat at artificially low prices on the international market. This is an example of the:

- A tariffs.
- B quotas.
- C exchange controls.
- D import licensing.
- E export subsidies.**
- F none of the above.
- G all of the above.

Question 13

Adam wants to use marginal utility theory to determine how many times per month he would like to visit the theatre. The following figure applies to his situation:



If he chooses the optimal amount, Adam will visit the theatre

A 4 times, and his total utility is 8.

B 4 times, and his total utility is 32.

C 10 times, and his total utility is 50.

D 6 times, and his total utility is 16.

E none of the above.

Question 14

The managing director of a monopoly firm is given the following data for a given output level:

Marginal revenue = £11, Marginal cost = £10

Average cost = £13, Average revenue = £15

To maximise profits, the firm should

A reduce price and increase output.

B reduce price and reduce output.

C increase price and increase output.

D increase price and reduce output.

E none of the above.

Question 15

If the demand for Good X increases (due to a rise in the income of consumers) and the supply of Good X increases (due to increased worker productivity), then the equilibrium price of Good X:

A will fall, and the equilibrium quantity may rise or fall.

B will fall, and the equilibrium quantity will rise.

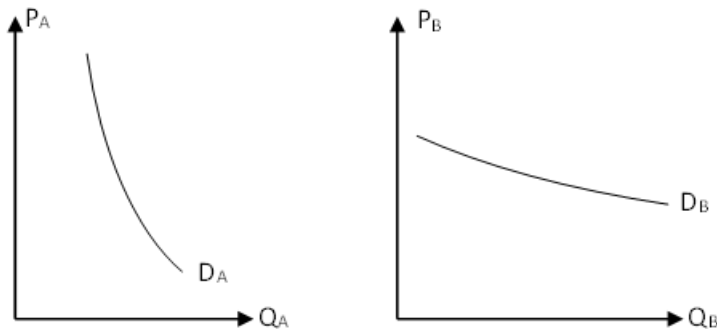
C may rise, fall or remain the same, and the equilibrium quantity will fall.

D may rise, fall or remain the same, and the equilibrium quantity will rise.

E none of the above.

Question 16

Consider the demand curves for product A (left panel) and product B (right panel), which are drawn using the same quantity scale and the same price scale.



Which of the following statements describes the demand for these two products?

- A The demand for A is more elastic than the demand for B, indicating that there are relatively few substitutes for A but several good substitutes for B.
- B The demand for A is less elastic than the demand for B, indicating that there are several good substitutes for A but relatively few substitutes for B.
- C The demand for A is more elastic than the demand for B, indicating that there are several good substitutes for A but relatively few substitutes for B.
- D The demand for A is less elastic than the demand for B, indicating that there are relatively few substitutes for A but several good substitutes for B.**
- E none of the above.

Question 17

Consumer A has a higher income than Consumer B, but they have identical preferences and pay the same prices for the goods they consume. If they both maximise utility, then

- A the marginal utility from each good consumed will be higher for A than for B, and A will have a higher total utility.
- B the marginal utility from each good consumed will be higher for A than for B, and A will have a lower total utility.
- C the marginal utility from each good consumed will be lower for A than for B, and A will have a higher total utility.**
- D the marginal utility from each good consumed will be lower for A than for B, and A will have a lower total utility.
- E none of the above.

Question 18

When a monopolist maximises profits, price exceeds marginal revenue. The difference between price and marginal revenue occurs because:

- A the firm has to charge a price higher than the marginal cost of producing the last unit.
- B any decision by the monopolist to sell an additional unit of output does not affect the price.
- C the firm has to reduce the price on all previous units sold in order to sell the additional unit.**
- D the law of diminishing returns directly affects the price of an imperfectly competitive firm's product.
- E none of the above.
- F all of the above.

Question 19

The short-run supply curve for a firm in a perfectly competitive industry is its

- A average total cost curve.
- B average variable cost curve.
- C marginal cost curve above the lowest point of the average total cost curve.
- D marginal cost curve above the lowest point of the average variable cost curve.**
- E none of the above.

Question 20

The following is a list of factors that can make trade beneficial. Into which one of these categories does the following example fit? Allowing imports freely into a country may stimulate domestic producers to be more efficient.

- A Decreasing costs.
- B Differences in demand.
- C Increased competition.**
- D Trade is an engine of growth.
- E none of the above.
- F all of the above.

Question 21

Assume that there is a free-floating exchange rate. UK interest rates rise relative to those abroad, then

- A the currency demand curve shifts right, the currency supply curve shifts left, and the exchange rate appreciates.**
- B the currency demand curve shifts left, the currency supply curve shifts right, and the exchange rate depreciates.
- C the currency demand curve shifts left, the currency supply curve shifts right, and the exchange rate appreciates.
- D the currency demand curve shifts right, the currency supply curve shifts left, and the exchange rate depreciates.
- E none of the above.

Question 22

As a result of falling unemployment, trade unions become more militant and demand higher wages. This will lead to

- A demand-pull inflation.
- B cost-push inflation.
- C both a and b.
- D neither a nor b.
- E none of the above.

Question 23

The success of having a single currency in Europe will depend on how close the eurozone is to an 'optimal currency area'. An optimal currency area can be defined as one which:

- A maximises the growth rates of the member countries.
- B minimises the degree of economic fluctuations between member countries.
- C maximises the amount of trade between the member countries.
- D minimises the average inflation rate between member countries.
- E would involve a decrease in net benefits from having a single currency if the size of the area were either to grow or diminish.**
- F none of the above.
- G all of the above.

Question 24

Assume that there are six firms in a country's carpet manufacturing industry and that they collude to restrict output, thereby maintaining a high price. Under which of the following circumstances is collusion likely to break down?

- A The government imposes higher duties on imported carpets.
- B Several new chains of discount carpet retailers enter the market.**
- C Three of the smaller manufacturers merge.
- D One of the firms becomes dominant in the industry.
- E none of the above.
- F all of the above.

Question 25

Assuming that interest rates and exchange rates are determined by the market, a reduction in money supply will result in

- A a fall in both interest rates and exchange rates.
- B a rise in both interest rates and exchange rates.**
- C a rise in interest rates and a fall in exchange rates.
- D a fall in interest rates and a rise in exchange rates.
- E none of the above.

Total: (50 marks)

Solutions Section B:

Question 26

The weekly demand and supply schedules for orange dresses (in millions) in a free market are as follows:

Price (£)	8	7	6	5	4	3	2	1
Quantity demanded	6	8	10	12	14	16	18	20
Quantity supplied	18	16	14	12	10	8	6	4

a) What is the equilibrium price and quantity?

(2 marks)

b) If a change in fashion causes demand for orange dresses to rise by 4 million at each price, what will be the new equilibrium? Has the equilibrium quantity risen by as much as the increase in demand? Explain why or why not.

(4 marks)

Answer

(a) Equilibrium is where the quantity demanded equals quantity supplied:

$$P = £5; Q = 12 \text{ million.}$$

(1 mark for each)

(b) The schedules following the increase in demand by 4 million units are shown in the following table:

Price (£)	8	7	6	5	4	3	2	1
<i>Quantity demanded</i>	10	12	14	16	18	20	22	24
<i>Quantity supplied</i>	18	16	14	12	10	8	6	4

Equilibrium price and quantity are now as follows:

$$P = £6; Q = 14 \text{ million}$$

(2 marks)

Demand has risen by 4 million, but equilibrium quantity has only risen by 2 million (from 12 million to 14 million). The quantity sold has risen by less than demand because the price has risen. This has choked off some of the extra demand (i.e. 2 of the 4 million).

(2 marks)

Question 27

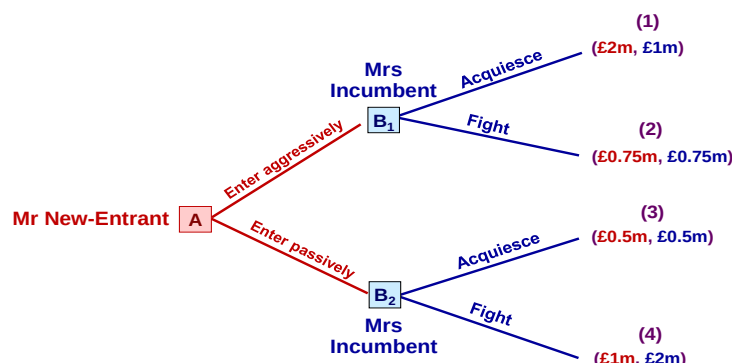
Consider the following sequential game: Mr New-Entrant is about to enter a market where there is an established firm run by Mrs Incumbent. Mr New-Entrant has to decide whether to enter the market aggressively (with deals and lots of advertising) or enter more passively. Once Mr New-Entrant has made his decision and entered the market, Mrs Incumbent has to decide between acquiescing (accepting that there is a new competitor) and fighting (trying to drive Mr New-Entrant out of the market.) If Mr New-Entrant decides to use aggressive tactics, he will make £2m profit if Mrs Incumbent acquiesces and £0.75m profit if she fights, while Mrs Incumbent would make £1m profit if she acquiesces and £0.75m if she fights. On the other hand, if Mr New-Entrant enters the market passively and Mrs Incumbent acquiesces, both will make £0.5m. If Mrs Incumbent chooses to fight despite the passive entrance, she will make £2m profit, while Mr New-Entrant will make £1m.

- Draw the game tree, clearly identifying who is the first mover, what each player's actions are and what their payoffs are.
(4 marks)
- What is the Nash equilibrium? Which Nash equilibrium do you think is the most likely to occur and why?
(4 marks)
- If Mrs Incumbent threatens that she will fight irrespective of the tactics Mr New-Entrant uses, should Mr New-Entrant believe that this threat is credible? Should this influence his decision?
(4 marks)

Answer

a) Mr. New-Entrant is the first mover, while Mrs. Incumbent is the second mover, who can observe what Mr. New-Entrant does and then best respond. Mr. New-Entrant has to decide between entering aggressively and entering passively and Mrs. Incumbent then has to decide under each potential scenario whether to acquiesce or fight.

This is illustrated in the following game tree. The payoffs are shown at the end of the game tree. Those in red are the payoffs of the first mover, Mr. New-Entrant, who enters at point A, and those in blue are the payoffs of the second mover, Mrs. Incumbent, who enters at point B₁ or B₂.



(2 marks)

(2 marks)

b) Nash equilibrium is where the optimal outcome is when there is no incentive for players to deviate from their initial strategy. The players have knowledge of their opponent's strategy and still will not deviate from their initial chosen strategies because it remains the optimal strategy for each player.

(2 marks)

The most likely Nash equilibrium is where Mr. New-Entrant chooses to enter aggressively and Mrs. Incumbent acquiesces (point (1)). Mr. New-Entrant can look down the game tree and make a prediction about how Mrs. Incumbent will behave. If he enters aggressively, he knows that Mrs. Incumbent will best respond by acquiescing, thus giving him £2m in profit (point (1)). If he enters passively, he knows that Mrs. Incumbent will best respond by fighting, which will give him just £1m in profit (point (4)). He would certainly prefer to have £2m in profit than £1m profit and hence the most likely Nash equilibrium is for Mr. New-Entrant to enter aggressively and for Mrs. Incumbent to acquiesce.

(2 marks)

(c) If Mrs. Incumbent threatens to fight no matter what Mr. New-Entrant does, this is not a credible threat. If Mr. New-Entrant chose to enter passively, then Mrs. Incumbent would certainly choose to fight, as this would yield £2m in profit (point (4)), which is better than the £0.5m in profit she would get if she acquiesced (point (3)). But, if Mr. New-Entrant choose to enter aggressively, Mrs. Incumbent would not follow through with her threat. After all, by fighting, she would only make £0.75m in profit (point (2)). She could make more profit if she simply acquiesced (£1m at point (1)). Therefore the threat that she will fight should be seen as non-credible and hence it should not influence Mr. New-Entrant's behaviour.

(4 marks)

Question 28

Imagine that two countries, Richland and Poorland, can produce just two goods, computers and coal. Assume that, for a given amount of land and capital, the output of these two products requires the following constant amounts of labour:

	Richland	Poorland
1 computer	2	4
100 tonnes of coal	4	5

Assume that each country has 20 million workers.

a) What is the opportunity cost of a computer in (i) Richland; (ii) Poorland?

(4 marks)

b) Which country has a comparative advantage in which product?

(2 marks)

c) Assume that trade now takes place and that 1 computer exchanges for 65 tonnes of coal. Both countries specialise completely in the product in which they have a comparative advantage. How much does each country produce of its respective product?

(4 marks)

d) The country producing computers sells 6 million domestically. How many does it export to the other country?

(2 marks)

e) How much coal does the other country consume?

(4 marks)

Answer

- (a) (i) 50 tonnes of coal (for every 2 workers diverted to computer production, 50 tonnes of coal are sacrificed).

(2 marks)

- (ii) 80 tonnes of coal (for every 4 workers diverted to computer production, 80 tonnes of coal are sacrificed).

(2 marks)

- (b) Richland has a comparative advantage in computers (it can produce them at a lower opportunity cost than can Poorland).

(1 marks)

Poorland has a comparative advantage in coal (it can produce it at a lower opportunity cost than can Richland).

(1 marks)

- (c) Richland: 10 million computers.

(2 marks)

Poorland: 400 million tonnes of coal.

(2 marks)

- (d) 4 million (i.e. 10 million – 6 million).

(2 marks)

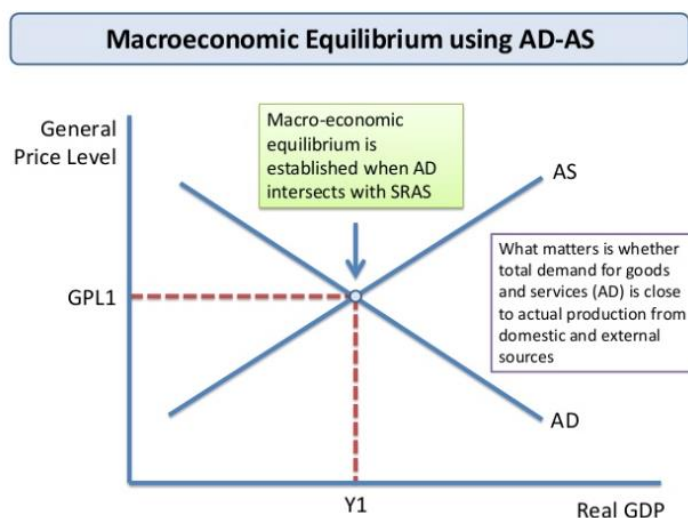
- (e) Poorland must import 4 million computers, for which it must export 260 million tonnes of coal (i.e. 4 million $\times$ 65 tonnes). It will therefore consume the remaining 140 million tonnes of coal.

(4 marks)

Question 29

With the aid of aggregate supply and demand curves, explain the demand-pull inflation.

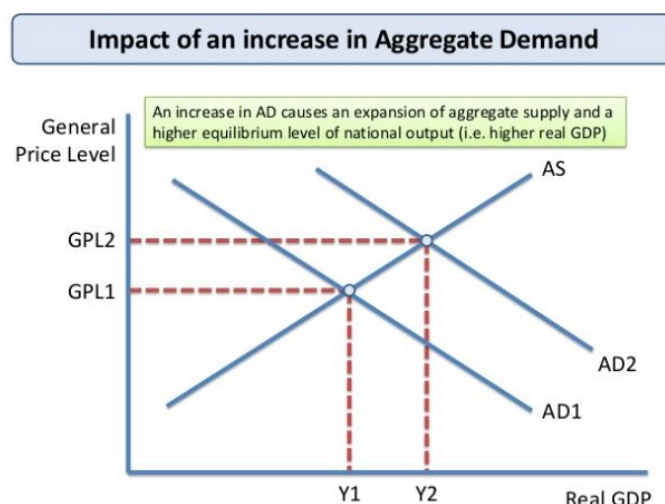
(8 marks)



Demand-pull inflation is a type of inflation that results from an initial increase in aggregate demand as shown in the diagram below. Demand-pull inflation may begin with any factor that increases aggregate demand:

1. Increase in the quantity of money and lower interest rates
2. An increase in government expenditures or tax cuts ($GDP = G + C + I + X$)
3. Changes in consumer and investment demand
4. An increase in export demand due for example to global economic growth

(4 marks)



(2 marks)

A shift to the right of the aggregate demand curve will lead to upward pressure on prices and some expansion of output. It is more likely to result in higher price rises closer to the full employment level of output than when there is a high level of unemployment.

(2 marks)

Question 30

Give the four subdivisions of a current account. For each, briefly explain what it records.

(8 marks)

Trade in goods account – records imports and exports of physical goods.

(2 marks)

Trade in service account – records imports and exports of services.

(2 marks)

Income flows – records wages, interest and profits flowing into and out of the country.

(2 marks)

Current transfers of money – records international contributions by governments and organisations and international transfers of money by private firms and individuals.

(2 marks)

Total: (50 marks)